PetCuesGame – Educational Animal Safety Game

Project Check-in

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PetCuesGame & Lecture Concepts

1. **User-Centered Design (UCD)**
- Child-friendly interface for ages 5-12.
- Web-based and accessible on PCs, tablets, and mobile devices.

2. **Agile Methodology**
- MVP Approach – Focus on core learning scenarios.
- Continuous Feedback from educators and the Regina Humane Society.

3. **Ethical Considerations**
- No personal data collection (COPPA & GDPR compliant).
- Open-source for global accessibility

Project Demo

Research Paper Progress

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*Note: Sub-titles are not captured in Xplore and should not be used

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***Abstract*—The PetCuesGame is an interactive, web-based educational tool designed to teach children safe and respectful interactions with animals. This project addresses a critical gap in traditional animal safety education, which often relies on brochures and in-person demonstrations that may lack engagement and long-term retention. By leveraging choice-based gameplay and interactive scenarios, the game provides an immersive learning experience where children (ages 5-12) can make decisions and receive real-time feedback on proper pet interaction behaviors.**

Aligned with the United Nations Sustainable Development Goals (SDGs), the project promotes Good Health Well-being (SDG 3) by reducing the risk of animal-related injuries, Quality Education (SDG 4) through an accessible, gamified learning platform, and Life on Land (SDG 15) by fostering empathy and responsible pet care. The game is built using the Godot Engine (GDScript) and is optimized for use on PCs, tablets, and mobile devices, ensuring widespread accessibility.

This paper outlines the technological framework, educational methodology, and the potential local and global impact of the PetCuesGame. By offering an engaging and scalable solution, the project not only supports the Regina Humane Society's mission of promoting responsible pet ownership but also sets the foundation for future enhancements, including multilingual support and expanded animal behavior scenarios. The paper discusses the challenges, solutions, and vision for a safer, more informed community through innovative digital education.

I. Introduction

Background – Why is animal safety education important?

Animal safety education is crucial in promoting safe and respectful interactions between humans and animals, particularly for children who frequently encounter pets in public spaces and at home. With the rise in pet ownership globally, the likelihood of children interacting with animals has increased significantly. While these interactions can foster compassion and emotional well-being, improper handling or misunderstanding of animal behavior can lead to injuries or stressful experiences for both the child and the animal. According to animal welfare organizations, a significant number of dog bites and related injuries occur due to a lack of knowledge on how to approach and engage with animals safely. Teaching children how to interpret animal body language and interact responsibly reduces the risk of incidents and helps create safer environments for both pets and people. Furthermore, providing children with effective animal safety education promotes empathy, responsibility, and compassion, fostering healthier relationships between humans and animals.

Problem Statement – Lack of structured, engaging education for safe pet interactions

Despite the importance of animal safety education, there is a lack of structured and engaging resources that effectively teach children how to interact with pets. Traditional educational methods, such as brochures, classroom lectures, and in-person demonstrations, often fail to capture young learners' attention or provide the hands-on, decision-making experience needed for long-term retention. This gap in education not only increases the likelihood of unsafe pet interactions but also limits the opportunity to cultivate responsible pet ownership from an early age. Current animal safety programs are largely passive and inaccessible to many communities, highlighting the need for an interactive, scalable, and engaging educational tool that can effectively teach animal safety practices.

Research Objectives – Educate children using an interactive, web-based approach

The primary objective of the PetCuesGame project is to develop an interactive, web-based educational game that teaches children (ages 5-12) safe and responsible pet interactions. This project addresses the current gap in animal safety education by offering real-world scenarios where players engage with virtual animals and receive immediate feedback on their decisions. The game is designed to:

- Promote Safe Animal Interactions – Teach children key behaviors such as asking permission before petting, rec-

Research Paper Progress

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Identify applicable funding agency here. If none, delete this.

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- Promote Safe Animal Interactions – Teach children key behaviors such as asking permission before petting, recognizing animal body language, and understanding safe approaches.
- Increase Engagement through Gamification – Use choice-based gameplay to make learning fun, interactive, and memorable.
- Ensure Accessibility and Scalability – Deliver the game through a web-based platform compatible with PCs, tablets, and mobile devices, allowing easy adoption in schools, humane societies, and home environments.
- Support United Nations Sustainable Development Goals (SDGs) – Align the project with SDG 3 (Good Health Well-being) by reducing animal-related injuries, SDG 4 (Quality Education) by providing an engaging educational tool, and SDG 15 (Life on Land) by promoting responsible pet care.

By leveraging technology and interactive learning, PetCuesGame aims to transform how animal safety education is delivered, making it accessible, engaging, and impactful for diverse communities. This paper explores the technological framework, educational benefits, and potential global impact of the project while addressing the challenges and future possibilities of improving animal safety education.

- Avoid combining SI and CGS units, such as current in amperes and magnetic field in oersteds. This often leads to confusion because equations do not balance dimensionally. If you must use mixed units, clearly state the units for each quantity that you use in an equation.
- Do not mix complete spellings and abbreviations of units: "Wb/m^2" or "webers per square meter", not "webers/m^2". Spell out units when they appear in text: ". . . a few henries", not ". . . a few H".
- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

C. *Equations*

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \quad (1)$$

Thank You